PAPER 3 - HANDOFF v14.0

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Written 8 September 2026. Supersedes v13.0 where they disagree.

Keep for history, not repeated in full: **v13.0** (hold-window bug, residual guard, calibration redo, real-robot collector, Real Robot tab), **v12.0** (grid thinning, throat jam, grid_quality / grid_accuracy / run_manifest, closing mode), **v11_2** (shape rules, scaled cuboid, shape-keyed calibration — internally titled “v13.0”; go by file name), **v10_0** (Block 2/3, diagonal blob), **v9_0** (oblique contact, motion pipeline).

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0. THE THIRTY-SECOND VERSION

Since v13.0 the work has been almost entirely in `design_grid`. A **transpose bug** was found that made every rolled-pad grid wrong, and fixing it properly meant replacing two independent limit calculations with one coupled solve valid at any pad roll and any object tilt. A second fault — an **unchecked across anchor** — was found immediately afterwards and closed, which retires the last item on the known-broken list.

Twelve upright sweeps are collected (6 cuboid, 6 cylinder). A **rolled 90° cylinder batch** is being started now, on five diameters.

Kourosh is on Paper 2 reviewer responses for the next two weeks. Nothing in this document is urgent. Block 3 remains the priority when he returns.

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1. THE OBJECTIVE (unchanged)

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- **Paper 1** (J. Robotics and Mechatronics 38(3), 2026) — shape classifier + hand-crafted extrapolation. 94.3% held-out, 80.8% unseen real. Safe-zone TC threshold 4.42 mm.
- **Paper 2** (Robotics and Autonomous Systems) — tactile-guided regrasping, CDT, 602 trials. **Reviewer comments received; response in progress.**
- **Paper 3** (this work) — replace the classifier and hand-crafted extrapolation with a **learned contact-completion model** trained on large-scale Isaac Sim data, then validate on the real rig.

Four blocks. Block 1 collection: complete. Block 2 stitching: complete and closed. Block 3 U-Net: **not started**. Block 4 A/B vs Paper 1: not started.

Stack. UR5e + Robotiq 2F-85, two TSF-85 pads (7×4 = 28 taxels each), Isaac Sim 5.1, TSF-85 extension by Berith Atemoztli De la Cruz Sánchez, cuRobo, Ubuntu 22.04. Real cell adds ROS 2 Humble, MoveIt, `ur_robot_driver`.

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2. WHERE THE PROJECT IS

Area	State
Block 2 stitching	Closed.
Calibration	7 keys, both files, one basis (deformation / 1.40 mm).
Upright cuboid sweeps	6 collected. **Not yet analysed.**
Upright cylinder sweeps	6 collected. **Not yet analysed.**
Rolled 90° cylinder sweeps	Batch starting now, 5 diameters.
`design_grid`	Rewritten for arbitrary roll and tilt. Anchor now owned by the designer.
Real robot	Motion, guards, home, shapes all working in free air.
Real tactile	**Blocked** — one pad missing from the lab.
Real calibration	**Not measured.** Every real run is `sim_fallback`.
Block 3	Not started.
Spheres	Untouched, deliberately.

Verify file freshness before editing

File	Signature
`main_gui.py`	`THE COUNTS ARE SOLVED TOGETHER` and `across_max_mm`
`viz/stitching.py`	`HOLD_PCTL`
`viz/grid_quality.py`	`FRAME_RATIO_MAX` / `FRAME_MIN`
`sim/collect_from_config.py`	`pad_to_pad_residual`
`examples/run_batch.py`	`GRASP_OBJECT_BOX_MM`
`Real_Robot/collect_real.py`	`home_vertical` / `arrival_residual` / `cal_key`
`viz/heatmaps.py`	`MIRROR_S2 = False` and `INDENT_MM = 1.4`

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3. THE ROLLED-GRID TRANSPOSE — THE MAIN FIND

3.1 What was wrong

Every limit in `design_grid` is stated in the **object’s** frame: the contact band constrains motion **across** the object, the palm and the ends constrain motion **along** it. But `_offsets` builds the lattice and then calls `rotate_offsets(offs, pad_roll_deg)`, which turns it into the **pad’s** frame *after* those limits have been applied.

At 0° the rotation is the identity, which is why every upright design ever made is correct and this survived unnoticed for months. **At 90° it transposes them**: the count derived from the along-window lands across the object, and vice versa.

Measured, on a Ø26 × 140 design at 90° roll: `across_max` was 23.03 mm and the sweep reached **±36.0 mm** across the object, while only ±18.0 mm of the ±36.0 mm along-window was used. The two spans were simply swapped. **42 of its 91 points could not touch the rod at all.** Nothing refused, the maps would have stitched, and the outer columns would just have been empty.

3.2 The general rule

A lattice of (n_a, n_l) steps of `step`, rotated by φ, reaches its extremes at the corner:

```
across = step · (nₐ·|cos φ| + nₗ·|sin φ|)
along  = step · (nₐ·|sin φ| + nₗ·|cos φ|)
```

The two counts are **coupled** at every angle except 0° and 90°, and no pair of independent ceilings can express that.

3.3 The fix

`design_grid` now solves both counts together: the largest point count whose corner satisfies all four limits — the across band, the along window, and both canvas axes — with ties going to the squarer grid. The search is small and exact.

φ is measured against the OBJECT, not the world. Tilting the rod rotates its across direction just as rolling the pad rotates the grid, and only the angle *between* them decides how the sweep projects. The canvas keeps the world roll, because the 96 mm canvas is world-axis aligned.

Regression: 52 combinations at 0° — six cylinder widths, six cuboids, a sphere, both lengths, overhang on and off — **all bit-identical** to the old arithmetic. Nothing upright changes.

Ø26 × 140 across roll:

roll	n_across	n_along	points
0°	2	3	35
30°	2	3	35
45°	2	3	35
75°	3	3	49
90°	4	3	63

Note: 90° gives *more* points than 0°, because the pad's long axis now lies across the rod where the tight band is, so the pad reaches further past the strip. The rolled dataset is therefore **denser than the upright one at the same step** — correct, but the two batches are not comparable point-for-point.

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4. THE ACROSS ANCHOR — THE SECOND FIND

Immediately after the transpose fix, six rolled cylinder designs were built with the y anchor left at **+28.42 mm** from an earlier auto-y experiment. Every report said ACCEPTED. Between **3 and 45 points per design could not touch the rod.**

The four scalar limits describe the sweep's **half-span**; nothing checked where its **centre** was. `z` had always been derived (the midpoint of the along window); `y` was the one anchor left typed, and that asymmetry is what let a stale number through.

Fixed. `design_grid` now owns both anchors:

- **Cylinder or sphere** → y is forced to 0, with a dialog naming the old value and why it was wrong. A rod has no side edge; the contact strip sits at its axis wherever the pad goes, so the only correct anchor is 0.
- **Cuboid or cube** → auto-y works exactly as before; a typed y is now **gated against `across_max`** and reset to the derived edge anchor if the outermost column would sit past what the pad can reach.

This retires **"No ACROSS-side limit on a hand-typed y"**, on the known-broken list since v12.0 §9.2.

The checkbox rules, restated

checkbox	upright cylinders	cuboids
auto y anchor	OFF (now ignored anyway)	ON
let the pad hang past the ENDS	ON	ON
centred grid	ON	OFF
step along pad axes	ON	ON
coarse interior	OFF	ON (N=2, band −11)

For **rolled** cylinders, auto-y and pad-hang make no difference: they relax scalar limits, but the binding constraint at 90° is the **measured gripper body**, which they do not touch. Verified on Ø72: with both on, the along window widened 44.08 → 63.08 mm and the trim went 6 → 8 steps, landing on the same 3 points.

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5. THE ROLLED 90° CYLINDER BATCH

Designed at y = 0, step 6 mm, no overhang, no auto-y, no thinning.

Ø	across_max	n_across × n_along	points	trimmed	worst rod-to-gripper gap
12	20.30	9 × 7	63	no	+32.96 mm
18	21.62	9 × 7	63	no	+22.41 mm
26	23.03	9 × 7	63	no	+16.09 mm
40	25.00	9 × 9	81	no	+4.88 mm
60	27.26	—	21	4 steps	+2.46 mm
72	28.42	—	9	6 steps	+4.55 mm

The trim on Ø60 and Ø72 is physics, not a fault. On a fat rod the surface falls away fast — at 24 mm off the axis a 36 mm radius has dropped 9.2 mm behind the tangent plane. The compliance band describes the *contact strip*, not the gripper **body**, and rolled 90° the pad's world-Y half-extent is 18.5 mm instead of 11.0, so the fingers reach the rod's shoulder long before the pad runs out of strip.

Decision: Ø72 was dropped from the rolled batch. Nine grasps over a 37 × 34 mm swept area carries almost no extension for a completion model to learn from. Ø60 at 21 points is marginal and was kept.

Note on the CNN: collecting rolled data at all is only defensible because 0° and 90° are the two angles Berith's CNN renders correctly (v9.0 §3.3: 0° reads −1.25°, 90° reads +89.7°). Intermediate angles still collapse into round blobs. `design_grid` now handles any angle correctly, but **the renderer does not** — do not collect 20°, 30° or 45° until the new extension passes its acceptance test (§8.1).

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6. THE REPORT WINDOW

`design_grid` used a `messagebox`, which does not scroll — the Ø72 diagnosis was unreadable. It now writes a full **design report**: a scrollable window with **Copy** and **Save to session folder**, and Save Config drops the same text beside `gui_config_used.json` automatically.

So from now on every run folder records **why its grid has the counts it has** — the band, the window, the coupling, the reach against each limit, and any point-check trim. The twelve upright sweeps predate this and have no report.

One bug of mine, now fixed: the report formatter referenced `self.vars['coarsen_interior']` (the *function* name) instead of `coarse_interior`, which threw a `KeyError` before the window was ever built. That is why the scroll and Save appeared not to exist.

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7. WHAT IS LEFT

7.1 Immediately on return

1. **Finish the rolled 90° cylinder batch** (5 diameters, in progress).
2. **Analyse all sweeps** — 6 upright cuboid, 6 upright cylinder, 5 rolled cylinder. Stitch → Grid Accuracy → Grid Quality → Save Run Manifest on each, then send `grid_quality_s1.txt` and `stitch_report.txt` as a set. Pad asymmetry like the Ø40 s1/s2 1.83 vs 1.35 mm is far easier to spot across widths than within one run.
3. **Block 3 — train the U-Net**, and nothing else. Pipeline exists (`dataset.py`, `model.py`, `train.py`) with masked loss, split-by-run, and baselines against zero and copy-input. **Use edge-straddling anchors, not pt00** — interior anchors halved TC error, 5.96 → 3.00 mm, crossing Paper 1's 4.42 mm threshold.
4. **Then Block 4** — A/B against Paper 1's shape rules.

7.2 Blocking real data (parallel, not on the critical path)

5. **Second tactile pad missing from the lab.** All real tactile numbers so far are meaningless — one run wrote all-zero maps across every taxel.
6. **No real-rig `TOOL_OFFSET_Z`.** Every real run is `sim_fallback`, putting a constant unverified offset into every real map. A real calibration procedure does not exist yet.
7. **Tactile frame rate.** 21 Hz on a freshly started Qt app, decaying to 1–3 Hz through a run — server load, not a fixed limit. At 1–3 Hz the hold window holds 5–10 frames against the sim's ~210.

7.3 Known-broken / not-yet-done

- `viz/validation.py` still assumes an axis-aligned pad footprint. **Correct on flat runs only — skip Validate on rolled runs.**
- **GSR is saturated.** Do not quote it.
- **A refused design does not block Save Config.**
- **Save Config must be pressed after Design grid.** Three configs were sent this session that still held the previous design. The report says the true count; the JSON says what was last saved.
- **Cuboid yaw is undefined.** Pads meet the +X/−X faces. Correct for a FACE grasp, but nothing records it, and an EDGE or corner grasp is a different signature Paper 1's rule does not cover.
- **`probe_finger_throat.py` models every object as a rod.** The design-time check uses the real box; the probed depth behind it does not.
- **`plot_scale.json` is `shared: true, fixed_vmax: null`** — each run scales to its own max, so brightness is not comparable across runs. Set 2400 (matching Paper 2) before pooling or putting two runs side by side.
- **The batch queue labels every entry "Ø30x140"** — `main_gui` writes the label from the grip width. Cosmetic; the folder path is what runs.
- **`--home-above-object` with `--object-here` OFF has never been run.** The real cell has **no collision world**, so IK will approve a path that sweeps through the table. Dry-run first with a typed object centre.
- **100 mm vs 140 mm bodies** — do not pool without saying so. One free-air real cuboid test used H=100.

7.4 Parked deliberately

- **Spheres.** Untested end to end. Berith's "spheres" are 40×40×90 and 65×65×90 capsules.
- **Intermediate pad rolls (20°, 30°, 45°).** `design_grid` handles them correctly now; the renderer does not.
- **Sequential tactile data.** The raw CSVs keep all ~250 frames per grasp permanently, so sequential pairs can be rebuilt from the same runs at any time without re-collecting. **Raise this at the start of Block 3** — it is a pair-construction decision, not a collection one.
- **A live tactile view for the real rig.** Use the Qt app; a second client on the same socket would compete for frames that are already scarce.

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8. CARRIED FORWARD

8.1 The diagonal blob artifact — still the biggest scientific limitation

Berith confirmed it is real and is working on it. 0° gives a clean vertical ridge and 90° a clean horizontal one; 20° and 45° collapse into broad blobs — at 45° the map has the **highest sum (12195)** with the **lowest peak (999)**. A completion model trained on maps where diagonal ridges render as round blobs would learn the renderer's failure, and no downstream metric would reveal it.

Berith is delivering a new TSF-85 Isaac Sim extension and later a Newton port. **Acceptance test, unchanged:** re-run 0/20/45/90 on the new extension and confirm 20° and 45° give elongated ridges before letting any intermediate-angle grasp into a training set.

8.2 Overlap sigma is still unexplained

Cylinder 46%, cuboid 36%. Squeeze variation was ruled out in v12.0 §7; collapsed hold windows were ruled out in v13.0 §4.4. **The row-gain artifact is the only candidate left standing.**

8.3 A small no-contact floor

Both sensors hold ~85 counts (7% of peak) at Y 221–225 on the Ø26 stitch, 9–13 mm outside the silhouette, after baseline subtraction. If you train on no-contact cells rather than masking them — which ShapeGrasp's free-space argument supports — that floor is what the model learns as "no contact".

8.4 A related paper worth citing

ShapeGrasp: Simultaneous Visuo-Haptic Shape Completion and Grasping, Rustler & Hoffmann, arXiv:2605.02347v2, June 2026 (unrefereed — cite as such). Three things to take: their free-space constraint argues for training on no-contact cells; their metric set (CD, HD, JS, F1, and the precision/recall split separating hallucinated from missing geometry) is sharper than SSIM alone for Block 4; and their reported failure mode is that tactile input *inflates* the completion — pre-register that as a check.

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9. CONSTANTS AND FORMULAS

Name	Value	Provenance
PALM_DROP_MM	86.69	measured: 10.79 + 75.9
THROAT_MARGIN_MM	1.0	bracketed
DEAD_GRASP_FLOOR	0.0002 m	rest 0.000, firm 1.3–1.5
CONTACT_TARGET	0.0014 m	uniform basis for every shape
across_margin_mm	3.0 mm	pad still touching at the outermost column
INDENT_MM (heatmaps)	1.4 mm	was 2.4; measured 1.16 on Ø26
indent_mm (coarsen)	1.4 mm	from the Ø26 measurement
HOLD_FRAC	0.9	unchanged
HOLD_PCTL	95.0	v13.0 §4
FRAME_RATIO_MAX / FRAME_MIN	2.0 / 50	healthy is ~210 frames
BLOCK_SHORTFALL_RAD	0.005	good points scatter 0.06 mrad
WEAK_DEFORM_FRAC	0.70	of the run's own median
canvas_mm	96.0	4 halvings; **world-axis aligned**
PAD_W, PAD_H	22.0, 37.0 mm	4 cols × 5.5, 7 rows × 5.29
OBJ_BASE_Z	982.2 mm	obj_z = 982.2 + L/2
grid step default	6.0 mm	0.5 mm sub-taxel shift per step
contact half-width	sqrt(2·R·delta)	compliance-limited, NOT silhouette
GRASP_MAX_RESIDUAL_MM	3.0	sim; now on BOTH routes
–arrival-tol-mm	1.0	real; arrivals run 0.002–0.169
–p2p-max-mm	25.0	real pad-to-pad ceiling
BASE_IN_WORLD_MM	[20.930, −337.500, 992.750]	measured 12 Aug, re-confirmed 5 Sept to 0.009 mm
real TOOL_OFFSET_Z	**unmeasured**	every real run is sim_fallback

Calibration keys (all 7, both files, deformation / 1.40)

key	TOOL_OFFSET_Z (contact)	close_rad	hold indentation
12.0	0.15717	0.6628	1.54 mm
18.0	0.15695	0.6079	1.45 mm
26.0	0.15649	0.5451	1.59 mm
40.0	0.15473	0.4097	1.83 mm
60.0	0.15048	0.2168	1.57 mm
72.0	0.14677	0.0930	1.44 mm
30.0|cuboid	0.15601	0.4994	1.72 mm

TOOL_OFFSET_Z spans 0.1468–0.1572 m: borrowing the wrong entry is worth up to 10 mm of pad position.

Object size plan

Cylinders, length 140 mm, spaced so the contact strip steps evenly (~2 mm), since the strip goes as sqrt(D):

Ø12 (strip 7.7, 35% of pad), Ø18 (9.6, 44%), Ø26 (11.7, 53%), Ø40 (14.7, 67%), Ø60 (18.1, 82%), Ø72 (19.9, 90%). Ø72 is the ceiling: at delta = 1.4 a cylinder needs Ø86 to fill the pad and the jaw opens 85.

Cuboids, grip 30 mm and along 140 mm fixed, across varied: 8, 14, 22, 34, 60, 120 mm. Grip is fixed because on a flat face the jaw gap changes nothing about the map — it is a nuisance parameter, and fixing it means one calibration and one throat probe cover all six.

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10. FILE INVENTORY — CHANGED SINCE v13.0

File	Change
`main_gui.py`	`design_grid` counts solved together for arbitrary roll and object tilt; `across_max_mm` returned; y anchor owned by the designer (forced 0 for round shapes, gated for flat); scrollable design report with Copy and Save; report written beside every saved config; `coarse_interior` key typo fixed

Unchanged since v13.0: `viz/stitching.py`, `viz/grid_quality.py`, `viz/grid_accuracy.py`, `viz/run_manifest.py`, `viz/heatmaps.py`, `viz/blob_axis.py`, `viz/validation.py`, `sim/collect_from_config.py`, `examples/run_batch.py`, `examples/probe_finger_throat.py`, `Real_Robot/collect_real.py`, `Data/build_object_library.py`.

The sweep sequence: Design grid → **Save Config** → run → Stitch → Grid Accuracy → Grid Quality → Save Run Manifest.

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11. HOW TO WORK ON THIS

- One verifiable step at a time. **Measure, don't guess.** Distances in mm.
- Complete files, never diffs. Commands on one line — multi-line `\` continuations silently drop env vars, and a prefix before `cd` does not survive `&&`.
- **One-line answers per question unless depth is asked for.**
- **Kourosh's visual read of the simulation overrides diagnostics when they conflict.** It has been correct every time.
- **Bad data must refuse LOUDLY rather than degrade silently.**
- Never present schematic pseudocode as a quotable line from an actual file.
- **Compute before asserting.** Twice this session a number was stated from memory and was wrong — a contact band given as 39.2 mm when it was 7.53, and a dead-point count that followed from it. The diagnosis survived; the arithmetic did not. Run it.

The pattern behind almost every fault in this project

A wrong number that looked normal. A hardcoded "cylinder" in a config. A grasped width taken as the minimum dimension. A calibration key two shapes shared. A gripper cloud filtered on the wrong axis. A ledger field that could not fail. A warning computed and then discarded. A guard on only one of two routes. A selection rule that discarded the data it was meant to average. A config knowing more than the code read.

Two added this session, and they rhyme with all of them:

- **Limits applied in one frame, then the geometry rotated into another.** The transpose was invisible at 0° and total at 90°.
- **A half-span checked and a centre not.** Every one of the six bad designs reported ACCEPTED.

None threw an error. **Every check prints WHAT IT MEASURED, not just its verdict — keep that up.**

End of handoff v14.0.